

Akshay Rajaram Patil

AI Engineer | Python Engineer | LLM and Automation

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PROFESSIONAL SUMMARY

AI Engineer and Python Engineer with experience designing, building and deploying production Large Language Model (LLM) systems and enterprise automation workflows. Delivered an autonomous incident resolution agent to production using Microsoft Copilot Studio, Microsoft Power Automate and Microsoft Dataverse. Built 100 plus Python tools integrated with LangChain and exposed through Model Context Protocol (MCP). Engineered 50 plus LLM personas with tuned decoding parameters. Skilled in Python, LangChain, LangGraph, Retrieval Augmented Generation (RAG), FastAPI, Django, REST API development, Prompt Engineering and SQL based data analysis.

TECHNICAL SKILLS

Artificial Intelligence and LLM Engineering: LangChain, LangGraph, Model Context Protocol (MCP), Retrieval Augmented Generation (RAG), Large Language Models (LLM), Generative AI, AI Agents, Agentic Workflows, Prompt Engineering, LLM Parameter Tuning, Temperature Tuning, Top-p, Top-k, Stop Sequences, Tool Schema Design, Vector Search

Programming Languages: Python, SQL, JavaScript, HTML, CSS

Frameworks and Libraries: Django, FastAPI, REST API, Pandas, NumPy, Matplotlib, JSON

Databases: MySQL, PostgreSQL, Microsoft Dataverse, SharePoint Lists

Automation and Cloud: Microsoft Power Automate, Microsoft Copilot Studio, Microsoft Power Platform, Workflow Automation, Microsoft Azure

Data and Analytics: Data Analysis, Data Cleaning, ETL, Power BI, Microsoft Excel, Data Visualization

Tools and Practices: Git, GitHub, Postman, API Testing, Jupyter Notebook, Visual Studio Code, Technical Documentation, Incident Management, Agile

PROFESSIONAL EXPERIENCE

AI and Automation Engineer (Graduate Trainee)

Tata Consultancy Services | Thane, Maharashtra, India | March 2026 to Present

- Designed, developed and deployed an autonomous incident resolution recommendation agent using Microsoft Copilot Studio and Microsoft Power Automate, owning the solution from architecture through production Go Live and earning direct client appreciation for design and development quality.
- Architected the end to end agentic workflow that classifies inbound incidents by mail trigger and route type, retrieves matching knowledge base articles, and returns recommended resolutions to support engineers.
- Led migration of the solution data layer from 4 SharePoint lists to Microsoft Dataverse, redesigning the relational schema and rebuilding all dependent Power Automate flows without production downtime.
- Developed and enhanced the AI Knowledge Block component enabling method level resolution classification and matching across knowledge base content.
- Improved production reliability by resolving save conflict errors, restructuring connection references and implementing concurrency safe apply to each handling following the data migration.
- Performed resolution accuracy analysis across large incident datasets using auditable, formula driven logic to identify inaccurate resolutions and unmatched knowledge base identifiers for management review.
- Authored technical documentation covering flow architecture, Dataverse schema design and step by step configuration, adopted as the reference standard for the solution.
- Received Star of the Month recognition for the successful production Go Live of the autonomous agent.

Data Science Intern (Python and LLM Engineering)

Smartinfoloigiks | Navi Mumbai, Maharashtra, India | July 2025 to January 2026

- Built 100 plus Python tools integrated with LangChain and exposed through Model Context Protocol (MCP), authoring structured tool schemas across multiple tool categories to ensure reliable model invocation.
- Engineered 50 plus LLM personas and skills, systematically tuning temperature, top-p, top-k and stop sequences to balance output quality, determinism and inference cost.

- Developed analytics dashboards on Analytics101, an internal JSON configured dashboard platform, authoring configuration schemas that drive chart rendering, filtering and layout behaviour.
- Delivered data retrieval for the Welspun client project, writing complex SQL queries against MySQL and implementing JSON parsing and transformation pipelines for downstream consumption.
- Standardised prompt engineering practices across production use cases, establishing reusable prompt structures and parameter configurations for consistent model behaviour.

AWARDS AND RECOGNITION

- Star of the Month, Tata Consultancy Services, 2026. Awarded for the production Go Live of the autonomous incident resolution agent.
- Runner Up, TCS AI Fridays Season 2, Mumbai, May 2026. Internal applied Artificial Intelligence and Generative AI competition.
- First Rank in Second Year and Third Year Bachelor of Science Computer Science. Third Rank in First Year.

EDUCATION

Bachelor of Science (B.Sc.), Computer Science

Karmveer Bhaurao Patil College, Vashi | Navi Mumbai, Maharashtra, India | 2022 to 2025

- CGPA 9.63, First Class.
- Relevant coursework: Data Structures, Database Management Systems, Python Programming, Web Technologies, Statistics.
- Active member, National Service Scheme (NSS), 1 year.

CERTIFICATIONS

- Django, The Complete Course 2026 (Beginner, Advanced and AI), Udemy, 2026.
- Python for Data Science, AI and Development, IBM through Coursera, 2024.
- Power BI, Prism Edu Skills, 2025.
- Python for Beginners, Infosys, 2022.

LANGUAGES

English (Professional Working), Hindi (Native), Marathi (Native)